

Impact of Data Breach on IT Investment: Embracing Both Failure Learning and Threat Rigidity

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Abstract

The extant literature has provided valuable insights into the post-failure behavior of organizations, highlighting two distinct tendencies: failure learning and threat rigidity. While failure learning involves organizations embracing change and seeking improvements after experiencing failures, threat rigidity leads to a more conservative and resistant approach to change during such times. In our study, we used a pioneering approach by integrating these seemingly competing perspectives within the context of data breaches. Employing a propensity score matching (PSM)-combined-difference-in-differences (DiD) approach, we uncovered a dual impact of data breaches on firms' information technology (IT) investment—after data breaches, firms tend to increase their IT investment intensity (a *promoting* effect) while simultaneously reducing their new IT investments (an *inhibiting* effect). Furthermore, we found that a firm with a strong quality culture exhibits a stronger tendency to increase its IT investment intensity following a data breach, while a firm highly valuing innovation demonstrates a weaker trend in reducing new IT investments after a breach. In post hoc analyses, we found that the impact of data breaches on IT investments is contingent on a series of factors related to the nature of the breach and the specific type of IT investments considered. Overall, our study provides valuable insights into the complex and diverse relationship between data breaches and IT investments in firms.

Keywords

Data Breach, Information Technology Investment, IT Innovation, Failure Learning Theory, Threat Rigidity Theory

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1 Introduction

In the age of ubiquitous computing, with information technology (IT) and electronic data permeating every aspect of business, data breaches have emerged as a pervasive operational challenge for firms. The frequency of data breach incidents has exhibited a rapid upward trajectory over the years. According to the 2023 annual data breach report, data compromises in 2023 surged by 78% compared to those in 2022, marking a new record for annual data breaches.¹ Moreover, data breaches carry substantial financial repercussions for firms. A recent study by the Ponemon Institute and IBM Security reveals that the average cost of data breaches in 2023 reached a record-breaking \$4.45 million.² In light of the heightened frequency and severity of data breaches, the importance of such incidents has been brought to the fore more than ever before, prompting inquiries into how data breaches might impact firms' strategic decision-making. In this study, we concentrated on a crucial element of corporate strategy in the digital era—the IT investment strategy of firms—and offer,

to our knowledge, the first consideration of how and why firms adjust their IT investment strategies in direct response to data breaches.

In the realm of IT investments following data breaches, existing evidence suggests the potential existence of two competing tendencies for firms. On the one hand, the “failure learning” perspective indicates that firms tend to learn from

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their failures and make adaptive investments to enhance future performance (Cyert and March, 1963; Gavetti et al., 2012). Taking Target, a well-known retailer, as a case in point, we noted that following a data breach, the firm significantly elevates its IT investments, with the objective of enhancing IT security measures and minimizing the potential for future data breaches.³ From this perspective, firms are likely to increase their IT investments after experiencing data breaches to fortify their information security and protect their stakeholders. On the other hand, the “threat rigidity” perspective proposes that failures can also lead firms to become more cautious and inflexible, in order to avoid potential unknown risks (D’Aunno and Sutton, 1992; Staw et al., 1981). For instance, a study by Accenture found that, within the banking sector, concerns about data breaches have led to reduced investments in innovative IT solutions, as firms opt for more conservative steps to mitigate risks.⁴ From this perspective, breached firms may consider reducing IT investments, especially in areas of novel IT solutions that involve significant uncertainty.

These contrasting views outlined above suggest that data breaches may act as a double-edged sword for firms’ IT investments: such incidents may either promote or inhibit these investment actions. This theoretical tension raises the intriguing questions: *do data breaches promote or inhibit firms’ IT investment? Could it be that data breaches have both promoting and inhibiting effects on different aspects of IT investment?* Despite the inherent significance, these questions have remained largely unexplored in existing research. To fill this knowledge gap, our study takes a comprehensive approach by examining the “failure learning” and “threat rigidity” perspectives simultaneously within the context of data breaches. The concurrent focus of such contrasting perspectives enables us to propose a *dual impact* of data breaches on IT investments. Specifically, from the “failure learning” perspective, we suggested that data breaches may act as catalysts, *prompting* firms to increase their IT investment intensity. On the other hand, drawing insights from the “threat rigidity” perspective, we proposed that breaches may also induce firms to adopt a cautious stance, *inhibiting* their adoption of new IT investments.

If firms’ data breaches indeed impact their IT investments, it is also crucial to understand how this effect may vary. Firms’ responses to failures, including data breaches, are intricately intertwined with their unique organizational characteristics, such as corporate culture, which serves as the backbone for conveying corporate values and principles (Detert et al., 2000; Rami and Gould, 2016). Therefore, another objective of this study is to investigate the moderating effects of corporate culture on the relationship between data breaches and IT investments. We particularly focused on the *quality* and *innovation* dimensions of corporate culture, as they may directly influence the strength of the “failure learning” and “threat rigidity” mechanisms, respectively. Specifically, a quality culture within a firm tends to foster comprehensive problem-solving after failures, thereby likely promoting “failure learning” post-breach (Detert et al., 2000). In contrast, an

innovation culture encourages fresh thinking and adaptability to failure, potentially reducing “threat rigidity” post-breach (Martins and Terblanche, 2003; Sadegh Sharifirad and Ataei, 2012). Therefore, we predicted that a strong quality culture will amplify the effect of data breaches on increasing IT investment intensity, while a robust innovation culture will mitigate the impact of data breaches on decreasing new IT investments.

We utilized a longitudinal dataset from US public firms covering a 7-year period (2012 to 2018) to validate our hypotheses. To assess the causal impact of data breaches on IT investments, we utilized a robust identification strategy—a propensity score matching-combined-difference-in-differences approach. This approach allows us to compare firms’ IT investments between the treatment and control groups (the first difference) and between the pre- and post-breach periods (the second difference). In addition, we employed the PSM approach to enhance comparability between the two groups. Our findings strongly support our hypotheses. We found that firms tend to intensify their IT investments following data breaches, indicating a *promoting* effect of data breaches on IT investments. We also observed a reduction in new IT investments post-breach, suggesting a concurrent *inhibiting* effect on IT investments. Moreover, we found that a strong quality culture within firms strengthens this promoting effect of data breaches, while a robust innovation culture weakens such an inhibiting effect.

In our post-hoc analysis, we delved into the nuances of how the impact of data breaches on IT investments may vary contingent on the severity of the breach. Large-scale breaches, which often result in more severe consequences such as significant financial losses, reputational damage, and compromised customer trust (Kamiya et al., 2021; Kwon and Johnson, 2015), tend to trigger stronger post-breach reactions among firms. As a result, the effects of both the “failure learning” and “threat rigidity” perspectives on IT investment are likely to be intensified. Our empirical findings support this expectation, revealing that the impacts of data breaches, whether promoting or inhibiting IT investments, are heightened in the aftermath of large-scale breaches.

Simultaneously, the impact of data breaches on IT investments can also vary depending on the nature and characteristics of these investments. Should data breaches induce firms to cut back on new IT investments due to threat rigidity, the extent of reduction should depend on the level of threats and uncertainties associated with the new IT investment. We recognized that the level of security threats associated with a new IT solution frequently depends on its security orientation and innovativeness, as these features play a role in shaping the security functionalities and risk resilience of IT (Angst et al., 2017; Wang et al., 2023). Therefore, to gain insight into the multifaceted relationship between data breaches and new IT investments, we took an additional step by categorizing new IT investments into two dimensions: non-security versus security types, and non-innovative versus innovative types. Our findings reveal that data breaches predominantly affect non-security-oriented or innovative new IT investments, with

security-oriented or non-innovative new investments experiencing comparatively less impact.

Altogether, we shed light on an important and dual-edged influence of data breaches on IT investment. This finding unveils, for the first time, that “failure learning” and “threat rigidity” could indeed coexist within the context of post-breach IT investments. Our findings also highlight that these post-breach responses are contingent upon various factors including corporate culture, breach characteristics, and the nature of IT investments involved. These insights underscore the complex interplay between these contextual factors and IT investment decisions in response to information security incidents. Hence, our study helps to further open the black box of how firms strategically respond to data breaches. Our findings have important implications, as we later discuss in the concluding remarks section of our paper.

2 Literature Review

Our study is closely related to two significant streams of literature. One is the literature stream on the consequences of data breaches. The other is the literature exploring the determinants of IT investments. These two streams of literature are reviewed as follows.

2.1 Consequences of Data Breaches

In the digital era, data have become one of the most critical components of an enterprise. A data breach occurs when confidential or private information is intentionally or inadvertently accessed by unauthorized parties (Cheng et al., 2017). With the exponential growth in data volume and the ever-increasing reliance on the Internet for business operations, data breaches have become more frequent than ever before, causing significant concern among executives. According to PwC’s 2020 survey, over half of the US chief executive officers (CEOs) have expressed extreme concern regarding the cyber threats their firms are facing, ranking such threats as the most significant danger to their organizations.⁵

Against such a backdrop, the past two decades have witnessed a proliferation of research exploring the consequences of data breaches. The earliest related studies investigated how data breaches influence firms’ market values (e.g., Campbell et al., 2003; Hovav and D’Arcy, 2003; Hovav and D’Arcy, 2004). However, the empirical findings of these studies are conflicting despite a solid theoretical basis for assuming an overall negative economic impact of data breach announcements. That is, while many studies have shown a statistically significant negative economic impact (e.g., Cavusoglu et al., 2004; Goel and Shawky, 2009), numerous other studies have found no significant results on the issue (e.g., Campbell et al., 2003; Hovav and D’Arcy, 2003; Kannan et al., 2007).

Several more recent studies have shifted their focus to firms’ operational performance and investigated the potential impact of data breaches in terms of related outcomes. For

instance, Hilary et al. (2016) conducted a study on all types of data breaches and found that attacked firms do not experience significant changes in their operational performance. Kwon and Johnson (2015) narrowed their investigation to the medical industry and discovered that the accumulation of data breaches may influence subsequent outpatient visits and admissions. Makridis and Dean (2018) explored the relationship between the number of breaches and firm productivity and identified a negative association, with the strength of the association contingent on the types of breaches involved.

The most recent focus has been directed toward firms’ strategic changes following data breaches. For instance, Benaroch and Chernobai (2017) conducted a study to investigate the influence of data breaches on firms’ board-level IT governance. They found that operational IT failures compel firms to enhance the IT competency level of their board members, especially if these failures trigger significant negative market reactions. Lending et al. (2018) explored the relationship between data breaches and corporate social responsibility (CSR) investments. They found that breached firms tend to increase their CSR investments to rebuild reputations. Furthermore, Nordlund (2019) and Say and Vasudeva (2020) discovered that breached firms are more likely to experience turnover in their organizational leadership and undertake unrelated divestitures.

Overall, the preceding literature review highlights the increasing academic interest in studying the consequences of data breaches, especially concerning firms’ strategic adjustments after experiencing such incidents. However, the current research landscape predominantly centers on firms’ adaptive enhancements following breaches, aligning with the “failure learning” perspective. However, it overlooks whether data breaches might also trigger firms’ heightened concerns regarding the relevant risks, potentially leading to behavioral rigidity. To address this research gap, we provided novel insights from a fresh perspective by introducing the contrasting concept of “threat rigidity.” Specifically, we explored the concurrent effects of “failure learning” and “threat rigidity” on firms’ IT investments after a data breach.

2.2 Determinants of IT Investments

In today’s digitalized and interconnected business landscape, IT plays a crucial role in corporate operations. Over the past two decades, extensive research has been dedicated to exploring the business value derived from IT investments. This body of literature aims to comprehend the influence of IT investments on various firm outcomes, including market reactions (Kohli et al., 2012), accounting performance (Bharadwaj et al., 1999), sales (De et al., 2010; Tan and Netessine, 2020), cost of capital (Liu et al., 2014), and labor productivity (Ding et al., 2010).

In contrast to the extensive body of work dedicated to quantifying the impact of IT investments, relatively limited

attention and research have focused on understanding the factors influencing firms' decisions regarding IT investments. Earlier studies explore how the internal characteristics of firms impact IT investment levels. Researchers such as Dewan et al. (1998), Harris and Katz (1991), and Mitra and Chaya (1996) delved into the influence of firm size on IT investment levels, yielding slightly varied empirical observations. Moreover, Dewan et al. (1998) and Hitt (1999) investigated the effect of firm structure on IT investments. Their consistent findings are that firms with more diversified and less vertically integrated structures often require higher IT investments to support decentralized information flow and complex collaboration among internal departments. Additionally, Xue et al. (2008) and Ravichandran and Liu (2011) explored the effects of IT governance factors, such as the degree of centralization and IT functional power, on firms' IT investment decisions. Their empirical findings underscore the crucial role these governance factors play in shaping such decisions. Furthermore, managerial characteristics are identified as additional factors influencing a firm's IT investment levels. For instance, Dong et al. (2021) found that a decline in performance stimulates IT investments, and reducing managerial agency problems can mitigate overinvestment in this process.

Previous research has also explored how a firm's external environment shapes its levels of IT investments. These studies suggest that market competition and institutional expectations can compel organizations to invest in IT. For example, Xue et al. (2008) and Ravichandran and Liu (2011) underscored the significance of environmental factors in organizations' strategic decisions regarding IT investments. Additionally, Ravichandran et al. (2009) found that a firm's IT investment intensity can be predicted by the IT investment intensity of its competitors, suppliers, and customers. Furthermore, Cheng et al. (2021) investigated the influence of interlocking directorates among firms on IT investments, highlighting that a focal firm's IT investments are affected by the interconnected levels of other board-interlocked firms.

In addition to variations in IT spending levels, firms differ in their eagerness to adopt new technologies. IT investments, especially in innovative technologies, come with substantial risks and often involve considerable knowledge barriers. Therefore, investing in these technologies, particularly for the first-mover advantage, entails associated risks (Jacobs et al., 2015; Tatikonda and Montoya-Weiss, 2001; Tatikonda and Rosenthal, 2000). Firms are inclined to invest actively in emerging technologies only when they possess the tolerance and capacity to bear such risks. This perspective has prompted studies examining factors influencing firms' orientations in new IT investments. For example, Xue et al. (2017) found that providing managers with risk incentives significantly shapes a firm's proactiveness in IT strategic posture. Barua and Kriebel (1991) determined that firms efficiently leveraging their IT investments are more likely to actively invest in IT. Grover and Goslar (1993) found that environmental uncertainty and decentralization of decision-making significantly influence the

adoption of new technologies. In addition, Chengalur-Smith and Duchessi (1999) found that a firm's market position, migration strategy, and system scope significantly influence the client-server technology initiation and adoption process.

As indicated by the above discussion, gaining a better understanding of how firms seek to leverage IT requires a multi-faceted exploration of their IT investment behavior. However, our literature review reveals that research in this area is not yet comprehensive. While some studies have linked firm characteristics, managerial features, or environmental factors to IT investment, there is limited attention to how specific firm events, such as firm failures, influence further IT decision-making. Nevertheless, such exploration is essential, as a substantial body of literature and various theories suggest that firm failures can trigger organizational adjustments and changes in decision-making (e.g., Gavetti et al., 2012; O'Brien and David, 2014). Along this line of reasoning, this study aims to address this gap in the existing literature by examining the impact of a specific IT operational failure, specifically data breaches, on a firm's IT investments.

3 Theoretical Background and Hypotheses

How do firms likely behave following failures? In answering this question, the perspective of failure learning stands against the one of threat rigidity. The former highlights that failures heighten the need for improvement and stimulate learning and changes (Cyert and March, 1963; Gavetti et al., 2012; Lattacher and Wdowiak, 2020; Thornhill and Amit, 2003), whereas the latter suggests that failures heighten awareness of danger and lead to rigidity and keeping still (D'aunno and Sutton, 1992; Staw et al., 1981). Therefore, although both perspectives explain how firms respond to failures, they offer competing predictions. In our study, we used an innovative approach by integrating the perspectives of failure learning and threat rigidity into a unified conceptual framework to investigate the impact of data breaches on IT investments. The remainder of this section elucidates the theoretical basis of our model and its hypothesized effects.

3.1 Data Breach and IT Investment: A Perspective of Failure Learning

As earlier mentioned, one line of thinking to address responses to failures is the failure-learning perspective, which is grounded in some theories, such as organizational learning theory (e.g., Levitt and March, 1988; Muehlfeld et al., 2012), prospect theory (e.g., Shimizu, 2007), the resource-based view (e.g., Thornhill and Amit, 2003), or behavioral theory of the firm (e.g., Cyert and March, 1963). Failure-learning theorists posit that firms' changes arise through a history of experience, particularly one with failures. Failures will evoke managers' concerns regarding firms' performance and motivate them to initiate a search for solutions. This ultimately leads to firms' reorientations, process refinements, and other major changes

and investments. Such a perspective of failure-learning resonates with voluminous literature on learning from failures (Haunschild and Rhee, 2004; Haunschild and Sullivan, 2002; Muehlfeld et al., 2012). Simply put, the failure learning perspective suggests that firms likely respond to failure by learning and making adaptive changes to pursue improvement and mitigate concerns stemming from failures.

Extending the perspective of failure learning to the context of data breaches, we argued that data breaches may prompt firms to escalate their IT investment intensity for several reasons. First, the necessity to restore regular operations frequently compels firms to increase their levels of IT investments after a breach. Specifically, data breaches frequently disrupt firms' business operations, impacting their ability to function effectively (Lohrmann and Tan, 2022; Wang et al., 2024). To restore functionality and bolster resilience, firms typically undertake a series of IT-related investments to assess and repair IT systems (Brown, 2016), such as isolating compromised systems, scrutinizing relevant network systems, reverting to backups, and rebuilding servers. In this context, significantly greater IT workforce and spending become imperative compared to normal situations (i.e., in the absence of data breaches).

Second, firms may increase their IT investments post-breach due to their heightened information security concerns. Data breaches often serve as a "wake-up call," abruptly highlighting the paramount importance of IT security (Masuch et al., 2022). This awareness may compel firms to adopt more proactive measures and invest in reinforcing their IT defense to strengthen their overall security posture (Gavetti et al., 2012). This perspective is substantiated by prior research indicating that firms affected by data breaches often reinforce their IT foundation post-breach, such as adjusting their IT governance structures and enhancing their IT competence (Benaroch and Chernobai, 2017). Such initiatives invariably require significant IT investment.

Third, firms may need to increase IT investments post-breach to restore customer confidence. Existing literature suggests that customers are deeply concerned about information security and are likely to adjust their consumption behaviors in response to information leakage events (Choi et al., 2016; Goode et al., 2017; Kwon and Johnson, 2015). Data breaches are anticipated to result in substantial customer loss for firms. According to IBM's 2019 cost of a data breach study, the erosion of customer trust constitutes a major cost category contributing to the overall cost of a data breach.⁶ Drawing on insights from the service recovery literature, customers often base their future consumption decisions on how firms address their data breaches (Hoffman and Kelley, 2000; Smith and Bolton, 2002). Therefore, to retain existing customers and rebuild customer relationships, breached firms may consider increasing IT investments as a means of enhancing their IT competency and demonstrating a high level of commitment to customers.

In summary, viewed from a failure learning perspective, managers are likely to acknowledge and learn from issues arising from data breaches, subsequently being inclined to increase their IT investments as a measure to address concerns. These insights resonate with existing research emphasizing that organizational failures often stem from resource weaknesses, and to mitigate vulnerabilities, firms need to invest in additional resources (Goldstein et al., 2011; Powell, 2001; Powell and Arregle, 2007). Therefore, we proposed the following hypothesis:

Hypothesis 1 (H1). Data breaches are associated with an increase in IT investment intensity.

3.2 Data Breach and IT Investment: A Perspective of Threat Rigidity

The perspective of threat rigidity offers a different way of understanding how firms respond to failures. According to this theory, when managers face failures, they may experience anxiety and tension, leading to conservative and rigid behaviors (D'anno and Sutton, 1992; Shi et al., 2018; Staw et al., 1981). This is because the pressure from failures reduces managers' inclination to process new information, causing them to focus on familiar solutions. Additionally, when a failure occurs, it often signals the risk of similar future failures, thereby causing managers to tighten control and formalize processes to prevent such failures from reoccurring. Generally, from a threat-rigidity viewpoint, following a failure, firms often exhibit a reduced propensity to explore new information and a tendency to maintain control, thereby leading their behavior to become more rigid and resistant to change.

When applying the threat rigidity perspective to the context of a data breach, we gained additional insights into the relationship between data breaches and IT investments that differ from those presented in Section 3.1. We argued that firms tend to reduce new IT investments following a data breach for the following reasons. First, data breaches may prompt firms to narrow their exploration of new IT options and thus reduce their new IT investments. When a data breach occurs, it exposes potential weaknesses and threats in the firm's IT infrastructure, leading to psychological stress and anxiety among managers regarding such threats (Fowler, 2016). The threat rigidity perspective suggests that in response to this stress and anxiety, managers may narrow their field of attention, simplify information codes, or reduce the number of channels used to process information (Shi et al., 2018; Staw et al., 1981). This can result in a limited exploration of new IT options and a focus on the existing IT portfolio, reducing the likelihood of adopting new IT investments. This perspective is supported by research that has found security threats can limit the number of alternatives considered by decision-makers (e.g., Holsti, 1964; Snyder and Paige, 1958). Therefore, following a data breach, due to threat rigidity, managers may reduce their efforts to explore new IT options, which, in turn, leads to a decrease in new IT investments.

Second, data breaches may prompt firms to enhance their control over existing information security operations and, compelling them to reduce new IT investments. Introducing new IT solutions often disrupts existing operations and introduces uncertainties and threats (Jacobs et al., 2015; Tatikonda and Montoya-Weiss, 2001; Tatikonda and Rosenthal, 2000), particularly in the information security domain. The reason is that new IT systems may lack well-established security protocols, procedures, policies, or norms, making it challenging to ensure favorable security outcomes (Mackelprang et al., 2015). Consequently, when firms invest in new IT, information security operations become relatively uncontrollable (Srinivas and Liang, 2022; Wang et al., 2023). From the threat rigidity perspective, firms' responses to breaches include avoiding the adoption of such uncontrollable changes and tightening control over proven operational aspects. Therefore, due to threat rigidity, breached firms may prioritize reinforcing their existing information security control and exhibit a propensity to reduce their IT investments in new domains.

Overall, when viewed from a threat rigidity perspective, data breaches may trigger threat rigidity among managers, manifested by their behaviors like curtailing IT exploration and tightening control, ultimately resulting in their decreased willingness to adopt new IT solutions. Hence, we hypothesized that:

Hypothesis 2 (H2). Data breaches are associated with a decrease in new IT investments.

3.3 The Moderating Role of Corporate Culture

If data breaches indeed impact a firm's IT investments, it is essential to understand how contextual factors may influence this impact. Examining contextual factors not only contributes to generalizing the identified main effects but also facilitates further testing of the core logic behind our hypotheses. In this section, we delved deeper into potential contextual factors that influence the main effects.

Firm responses to failures are intricately linked to their contextual environment, with corporate culture being a pivotal aspect. Corporate culture serves as the underlying fabric that shapes the values, norms, and behavioral patterns within an organization (O'reilly and Chatman, 1996). It can foster different perceptions and attitudes toward failure, thereby significantly influencing how firms respond to their setbacks (Detert et al., 2000). Therefore, in this study, we aimed to explore the moderating effects of corporate culture. Further recognizing the multifaceted nature of corporate culture, we specifically focused on its quality and innovation dimensions, which, as discussed below, may directly influence the proposed "failure learning" or the "threat rigidity" mechanism.

3.3.1 The Moderating Role of Quality Culture: Shaping "Failure Learning". Quality culture refers to the inherent organizational values and practices that focus on maintaining high standards

and continuous improvement (Malhi, 2013; Sashkin and Kiser, 1993). Within a quality culture, firms tend to prioritize failures and mistakes, viewing them as crucial learning opportunities. Moreover, within this culture, firms are motivated to dedicate sincere efforts to thoroughly addressing the setbacks encountered in their operations (Cameron and Sine, 1999; Malhi, 2013; Sashkin and Kiser, 1993). As noted by Cameron and Sine (1999, p. 12), firms that value quality culture tend to "emphasize inspecting, in order to detect errors; avoiding mistakes; reducing waste; and finding and fixing defects," to adhere to their steadfast commitment to quality.

Hence, when faced with data breaches, these firms are inclined to place greater importance on such failures and prioritize learning from them. Throughout the process, they tend to adopt comprehensive strategies and adopt a responsible approach to problem-solving, demonstrating their steadfast commitment to secure operations. For example, they may conduct thorough root-cause analyses of the breaches and allocate extra resources to extract insights aimed at averting future setbacks. Put differently, firms with a robust quality culture are more likely to display a heightened tendency toward "failure learning" after experiencing data breaches. Given that our proposition of the promoting effect on increasing IT investment intensity post-breach is rooted in "failure learning," we anticipated that this effect would be further strengthened by a strong quality culture. Therefore, we proposed:

Hypothesis 3 (H3). A robust quality culture strengthens the positive association between data breaches and IT investment intensity.

3.3.2 The Moderating Role of Innovation Culture: Shaping "Threat Rigidity". Innovation culture represents an additional framework of values and behavioral patterns ingrained throughout an organization, emphasizing the importance of embracing new ideas and approaches. Within an innovative culture, firms tend to acknowledge and embrace failure more readily, understanding that innovation often involves navigating uncertainty and risk. As Coffman (1997) described, such firms will recognize that "there is no guarantee that experiments will run without unpleasant consequences; however, this is part of life" (p. 10) and they understand that "an innovator never sets out to fail" (p. 10). This positive attitude toward setbacks reduces their fear of failures like data breaches, fostering enhanced resilience and adaptability in addressing these challenges rather than being constrained by fear of change. Consequently, it is reasonable to expect that within an innovation culture, firms exhibit a lower level of "threat rigidity" in the face of data breaches. According to our theoretical framework, the inhibiting effect of data breaches on reducing their new IT investments is driven by "threat rigidity," and consequently, we predicted that a robust innovation culture tends to alleviate such an effect. Accordingly, we proposed:

Hypothesis 4 (H4). A robust innovation culture weakens the negative association between data breaches and new IT investment.

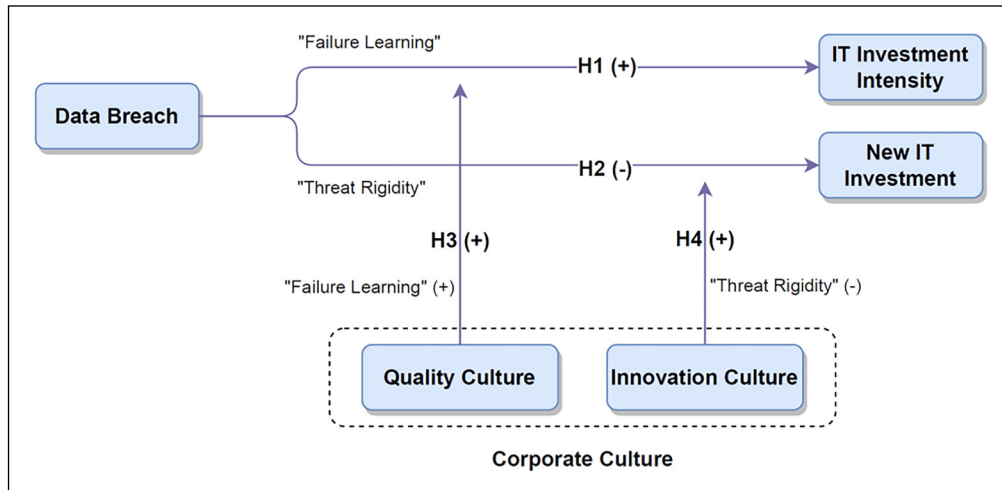


Figure 1. Conceptual framework.

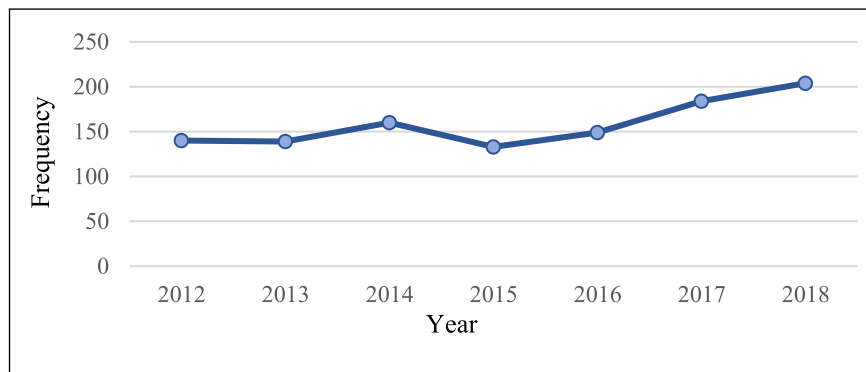


Figure 2. Year distributions of breaches in the data breached sample.

Figure 1 illustrates the conceptual framework developed for this research.

4 Sample Selection and Methodology

4.1 Data and Sample Selection

Our study focuses on publicly traded firms in the US. To comprehensively collect data on data breaches, we utilized three data sources: (1) Wharton Research Data Services (WRDS) for cybersecurity data breach data,⁷ (2) Privacy Rights Clearinghouse (PRC),⁸ and (3) Identity Theft Resource Center (ITRC).⁹ If a publicly traded US firm has been recorded as experiencing a data breach in any of the mentioned data sources within our sample period (2012–2018), that breach event is included in our sample. Subsequently, we identified the stock ticker information for each breached firm included in the sample.¹⁰ Through the process, we obtained a total of 1,109 data breach events involving 648 publicly traded firms in the US over a 7-year period (2012 to 2018).¹¹ Figure 2 provides insights into the year distribution of these breaches, showing a fluctuating increase in data breaches over the sample period.

We collected data on IT investment for publicly traded firms in the US from Harte-Hanks' Computer Intelligence (CI) technology database. The CI technology database contains annual IT infrastructure information for over 500,000 sites in the US and Canada. It is widely regarded as a reliable source of firm-level IT data and has been extensively used in previous research studies (Chwelos et al., 2010; Kleis et al., 2012; Tian and Xu, 2015; Xue et al., 2011; Xue et al., 2013; Xue et al., 2017). Our analysis focuses on the data related to IT spending and IT implementations recorded in the CI technology database during our sample period, which spans from 2012 to 2018.

After integrating the data from the mentioned sources with financial information from Compustat, we established the final full sample. To address the impact of outliers in our subsequent tests, we applied winsorization to all firm-year continuous variables at the 1% and 99% levels, respectively. The full sample consists of 17,832 firm-year observations, encompassing 2,648 unique firms. Moreover, the sample includes 631 data breaches involving 280 distinct firms during the period from

2012 to 2018. Table EC-1 (E-companion) presents the industry distribution in the final sample.

4.2 Operationalization of Variables

4.2.1 Outcome Variables. Our study primarily focuses on two outcome variables: IT investment intensity (*Intensity*) and new IT investment (*Innovation*). These variables are derived from data obtained from the CI technology database. First, we operationalized *Intensity* as the size-scaled sum of IT expenditures across all units in a firm. Due to significant skewness in the original values, a natural logarithm transformation is performed. Second, we constructed *Innovation* as the count of new IT adoptions by a firm in a specific year. We identified a new IT adoption if an IT was not implemented by a firm in the last year but is implemented in the current year. Specifically, we considered all 52 types of IT implementations recorded in the CI technology database for this construction.¹² We presented a detailed list of the applied IT implementations in the E-companion (EC-2).

4.2.2 Moderating Variables. In the main analysis, we examined two moderating variables: *Quality Culture* and *Innovation Culture*. Li et al. (2021) utilized machine learning to construct quality and innovation culture scores for firms.¹³ In this study, we directly adopted these scores for *Quality Culture* and *Innovation Culture*. A more detailed explanation of this measurement can be found in the E-companion (EC-3).

4.2.3 Controls. We have included a series of control variables that could potentially influence a firm's IT investments. First, we controlled the *Firm Size*, given that large firms often exhibit unique IT investment tendencies to meet extensive operational needs. Second, we accounted for the *Firm Age*, recognizing that younger firms might adopt more IT innovations to maintain competitiveness. Third, we incorporated return on assets (*ROA*) as a control variable, given that highly profitable firms typically have more funds available to support IT investments. Fourth, we controlled for *Firm Loss Position*, considering that poor performance may stimulate firms to strategically accept IT investments as a means of recovery. Fifth, we controlled for research and development intensity (*R&D*), as research-intensive firms may rely more heavily on advanced IT to support innovation activities. Sixth, we included *Resource Slack* as a control, given that resource-rich firms possess greater funds and capabilities to support IT investments. Seventh, we incorporated *Digital Connectivity Scope* as a control, given that extensive remote and mobile work in firms necessitates intensive IT investments. Finally, we controlled for the *Network Infrastructure Scale*, given that such a scale can directly affect the IT investments required by firms. For detailed definitions and data sources for all variables, please refer to Table EC-4 in the E-companion.

4.3 Identification Strategy

Our empirical analysis aims to identify changes in IT investment tendencies associated with data breaches. As only a subset of firms experienced data breaches, this provides an ideal setting to assess the impact of data breaches using the DiD method, which helps mitigate unobserved confounding factors. However, effectively estimating the impact of data breaches using the DiD method remains challenging due to the non-random occurrence of data breaches. Instead, various factors, such as market conditions and executive characteristics, may simultaneously influence both the likelihood of experiencing a data breach and IT investment decisions in firms. In such cases, data breaches and the associated changes in IT investments may only be correlated rather than representing a causal relationship.

To address this issue, we employed the PSM approach (Rosenbaum and Rubin, 1983) to help create a quasi-experimental environment (Rosenbaum and Rubin, 1983). Specifically, we applied the PSM approach to our full sample consisting of 17,832 firm-years. We categorized firms that experienced at least one data breach as the treatment group, while firms that did not experience any data breaches during our sample period constitute the control group. We ran a logit regression on a dummy variable, which equals 1 if a particular firm year belongs to our treatment group, and 0 otherwise. The regression includes all control variables and industry dummies as matching criteria. After estimating the logit model, we used the propensity scores obtained from this logit estimation and perform 1:1 nearest neighbor matching (without replacement). This process results in a 1:1 matched treatment and control group, each containing 1,958 observations. Therefore, the total PSM-matched sample includes 3,916 observations.

For brevity, we reported additional details of the PSM approach in Table EC-5 (E-companion). Panel A shows a comparison of the characteristics of breached and non-breached firms before the first data breach event. The results indicate that before data breaches, the breached and non-breached firms exhibited similar characteristics across all control variables, suggesting that these features are unlikely to drive differences in IT investments following a data breach. We also examined the differences in propensity scores between treatment firms and matched control firms. Panel B shows that the differences are substantially trivial. Overall, the diagnostic tests reported in Table EC-5 (E-companion) indicate that the PSM process helps eliminate apparent sample selection bias and reduces the likelihood that IT investment changes are caused by exogenous data breaches. Additionally, Figure EC-5 (E-companion) visually depicts the distribution of propensity scores before and after PSM matching, demonstrating that the differences between the control and treatment groups prior to matching have been substantially reduced after PSM. This further confirms the effectiveness of the PSM approach in creating balanced comparison groups.

Due to the staggered adoption of data breaches by different firms, we utilized a staggered DiD research design with a two-way fixed effects model (Dai et al., 2020; Schmidheiny and Siegloch, 2023). Specifically, we implemented the following DiD specification:

$$Outcome_{i,j} = \beta_0 + \beta_1 Post\ Breach_{i,j} + \sum \alpha Control_{i,j} + v_i + w_j + \epsilon_{ij}. \quad (1)$$

In Equation (1), i and j represent the firm and year, respectively. The outcome variable, *Outcome*, represents IT investment intensity (*Intensity*) or new IT investment (*Innovation*). *Post Breach* is a treatment indicator taking a value of 1 for a firm in the period after its first data breach and 0 otherwise. The main coefficient of interest is β_1 . To support H1, we expected β_1 to be positive when the outcome variable was *Intensity*, indicating a positive relationship between the post-breach status and IT investment intensity; to support H2, we expected β_1 to be negative when the outcome variable was *Innovation*, suggesting a negative relationship between the post-breach status and new IT investments. *Control* represents a series of control variables that we considered, and α denotes the corresponding estimated coefficient. Furthermore, v represents firm fixed effects, controlling for all time-invariant variables that could potentially influence IT investments, while w represents year-fixed effects, controlling for any time-related fluctuations in IT investment that are common across all firms in the sample.

5 Empirical Results

After merging data from multiple sources and applying the PSM approach, we obtained a PSM-matched sample comprising 3,916 firm-year observations from 2012 to 2018. Our subsequent DiD analyses will be based on this sample.

5.1 Descriptive Statistics

Table EC-6 (E-companion) presents descriptive statistics for the variables used in the subsequent DiD analysis across three distinct groups: (1) treatment firms in the pre-breach stage, comprising 1,089 firm-years, (2) treatment firms in the post-breach stage, consisting of 869 firm-years, and (3) control firms, totaling 1,958 firm-years. Table EC-6 reveals that around 22.2% of the firm-years in the PSM-matched sample are impacted by data breaches.

Table EC-6 also provides model-independent estimates of firms' post-breach IT investment tendencies. Specifically, in terms of IT investment intensity (*Intensity*), the average *Intensity* in treatment firms before a breach is 14.975, significantly increasing to 16.117 in treatment firms after a breach. In contrast, control firms have an average *Intensity* of 14.242. These statistics collectively indicate a promoting and increasing effect of data breaches on firms' IT investments. Moving on to new IT investment (*Innovation*), this level among treatment firms prior to a breach is 0.684 but significantly decreases to 0.338 among treatment firms after a breach. In contrast, control firms have an average *Innovation* of 0.436. These statistics

collectively imply an inhibiting and decreasing effect of data breaches on firms' IT investment.

5.2 Data Breaches and IT Investment (Testing for H1 and H2)

We began by testing H1 and H2, which are hypotheses pertaining to the impact of data breaches on firms' IT investments. Table 1 presents the outcomes of our baseline DiD regression analysis, utilizing the PSM-matched sample. In each model, we incorporated firm and year fixed effects.

In column (1) (Table 1), the dependent variable is *Intensity*. Here, we included only the treatment dummy, *Post Breach*, as an independent variable. The estimated coefficient of the term is 0.099, significant at the 5% level, indicating that IT investment intensity tends to increase after data breaches compared to the control group. Hence, we found supporting evidence for H1.

Table 1. Impact of data breaches on IT investment.

Dependent variable: For testing	<i>Intensity</i> H1		<i>Innovation</i> H2	
	(1)	(2)	(3)	(4)
<i>Post breach</i>	0.099** (2.208)	0.090** (2.024)	-0.185** (-2.035)	-0.202** (-2.228)
<i>Firm size</i>		0.315*** (5.664)		0.542*** (4.790)
<i>Firm age</i>		0.345*** (3.748)		0.191 (0.688)
<i>Firm ROA</i>		-0.442 (-1.111)		-0.818 (-1.013)
<i>Firm loss position</i>		0.008 (0.134)		-0.104 (-0.837)
<i>R&D</i>		2.569* (1.788)		-2.225 (-0.763)
<i>Resource Slack</i>		-0.044 (-0.908)		-0.169 (-1.612)
<i>Digital connectivity scope</i>		-0.000 (-0.728)		-0.000 (-1.605)
<i>Network infrastructure scale</i>		0.001 (1.101)		0.002* (1.727)
<i>Intercept</i>	Yes	Yes	Yes	Yes
<i>Firm fixed effects</i>	Yes	Yes	Yes	Yes
<i>Year fixed effects</i>	Yes	Yes	Yes	Yes
<i>N of observations</i>	3,411	3,411	3,200	3,200
<i>Adjusted R²</i>	0.872	0.874	0.123	0.130

Notes: This table presents coefficients from DiD regressions of the impact of data breaches on IT investments. The sample is a PSM-matched sample, which includes 3,916 observations. Our analysis employs the model specified in Equation (1) and utilizes Ordinary Least Squares (OLS) regressions. In models 1 and 2, we excluded 44 observations due to missing values for *Intensity* and 461 singleton observations. In models 3 and 4, we excluded 320 observations due to missing values for *Innovation* and 396 singleton observations. The t-statistics are reported in parentheses. The symbols ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

In column (2) (Table 1), we further introduced all control variables, obtaining estimates consistent with Model 1. In this model, the coefficient for *Post Breach* remains significant at the 5% level. The coefficient is also economically significant. According to this model, following a data breach, treatment firms would experience an average 9.4% increase in their IT investment intensity compared with the control firms.¹⁴

In column (3) (Table 1), the dependent variable is *Innovation*. Similar to Model 1, we included only *Post Breach* as an independent variable. The estimated coefficient of the term is -0.185 , significant at the 5% level, indicating a decreasing tendency in new IT investments after data breaches compared to the control group. Thus, we obtained evidence in support of H2.

In column (4) (Table 1), we further introduced all control variables and obtained consistent estimates with Model 3. In this model, the coefficient for *Post Breach* remains significant at the 5% level. The coefficient is also economically significant, revealing that following a data breach, treatment firms would experience an average 37.9% decrease in their average number of new IT adoptions compared to the control firms.¹⁵

Overall, we observed a dual effect of data breaches on firms' IT investment—a promoting effect (specifically, increasing IT investment intensity) and an inhibiting effect (specifically, decreasing new IT investment).

5.3 The Moderating Role of Corporate Culture (Testing for H3 and H4)

Subsequently, we examined H3 and H4, which are hypotheses pertaining to the moderating effects of corporate culture. Table 2 presents the relevant regression coefficients, utilizing the PSM-matched sample. In each model, we incorporated firm and year fixed effects.

To assess the moderating effect of *Quality Culture*, we introduced this variable and its interaction with *Post Breach* into Equation (1). In column (1) (Table 2), without introducing any control variables, the coefficient for *Post Breach* $\times$ *Quality Culture* is 0.059, significant at the 5% level. This indicates that when breached firms have a strong quality culture, the positive impact of data breaches on IT investment intensity is strengthened, thereby supporting H3. In column (2) (Table 2), we further introduced all control variables, and the results are consistent with column (1), thereby further reinforcing the robustness of the previous finding. Furthermore, we employed marginal effect plots to illustrate the moderating effect of *Quality Culture* (Brambor et al., 2006). Figure 3, column (a), shows that with an increase in *Quality Culture*, the positive marginal effect of data breaches on IT investment intensity gradually strengthens. This graphical representation aligns with the findings obtained from the regression models and provides additional support for the strengthening moderating effects of quality culture on the positive relationship between data breaches and IT investment intensity.

Table 2. Moderating role of corporate culture.

Dependent variable: For testing	Intensity H3		Innovation H4	
	(1)	(2)	(3)	(4)
<i>Post breach</i>	−0.025 (−0.324)	−0.041 (−0.542)	−0.438*** (−2.581)	−0.489*** (−2.884)
<i>Post Breach</i> $\times$ <i>quality culture</i>	0.059** (2.260)	0.064** (2.434)		
<i>Post breach</i> $\times$ <i>innovation culture</i>			0.047** (2.024)	0.052** (2.244)
<i>Quality culture</i>	0.019 (0.864)	0.021 (0.950)		
<i>Innovation culture</i>			−0.039* (−1.700)	−0.049** (−2.123)
<i>Intercept</i>	Yes	Yes	Yes	Yes
<i>Controls</i>	No	Yes	No	Yes
<i>Firm fixed effects</i>	Yes	Yes	Yes	Yes
<i>Year fixed effects</i>	Yes	Yes	Yes	Yes
<i>N of observations</i>	2,941	2,941	2,824	2,824
<i>Adjusted R²</i>	0.871	0.874	0.110	0.123

Note: This table presents coefficients from DiD regressions of the moderating effect of corporate culture. The sample is a PSM-matched sample, which includes 3,916 observations. Our analysis employs the model specified in Equation (1) by additionally introducing the interaction term between *Post Breach* and a certain corporate culture variable. We utilized OLS regressions. The t-statistics are reported in parentheses. The symbols ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

To assess the moderating effect of *Innovation Culture*, we introduced this variable and its interaction term with *Post Breach* into Equation (1). In column (3) (Table 2), the coefficient for *Post Breach* $\times$ *Innovation Culture* is 0.047, significant at the 1% level. This indicates that the decreasing effect of data breaches on new IT investments is weakened when breached firms possess a strong innovation culture, thus supporting H4. In column (4) (Table 2), we further introduced all control variables and found consistent results with the ones in column (2). We also used marginal effect plots to visualize the moderating effect of *Innovation Culture*. Figure 3, column (b), visually demonstrates the moderating effect of *Innovation Culture* by using marginal effect plots. This figure illustrates that *Innovation Culture* attenuates the negative marginal effect of data breaches on investments in new IT domains. This graphical representation aligns with the findings from our regression models and provides evidence supporting the weakening moderating effects of innovation culture on the negative relationship between data breaches and new IT investment.

5.4 Validations and Robustness Checks

In this section, we validated our research design and perform a series of robustness tests.

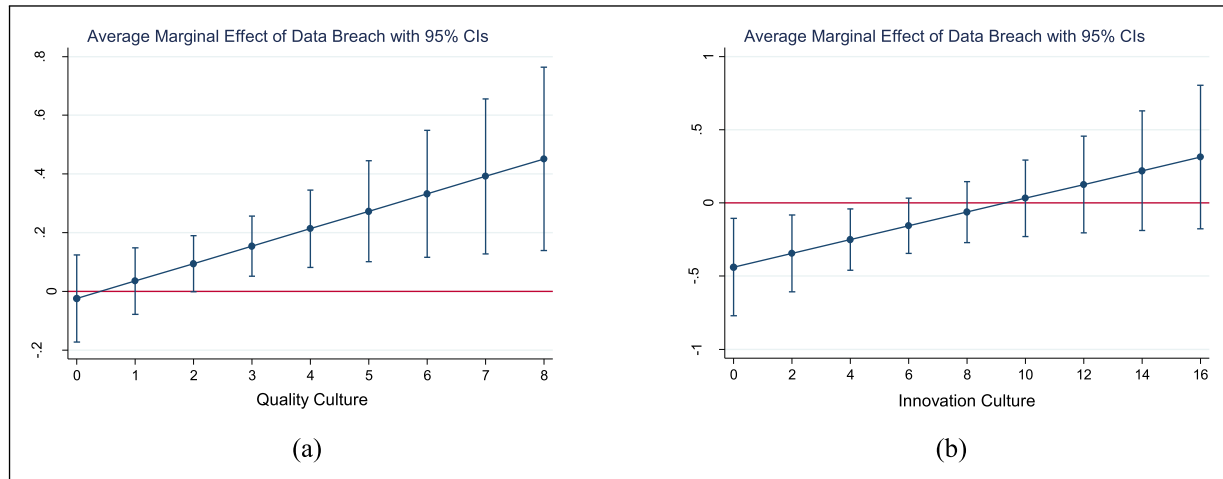


Figure 3. Marginal effect plot: (a) marginal effects of a data breach on the IT investment intensity and quality culture; (b) marginal effects of a data breach on IT investment in a new domain and innovation culture.

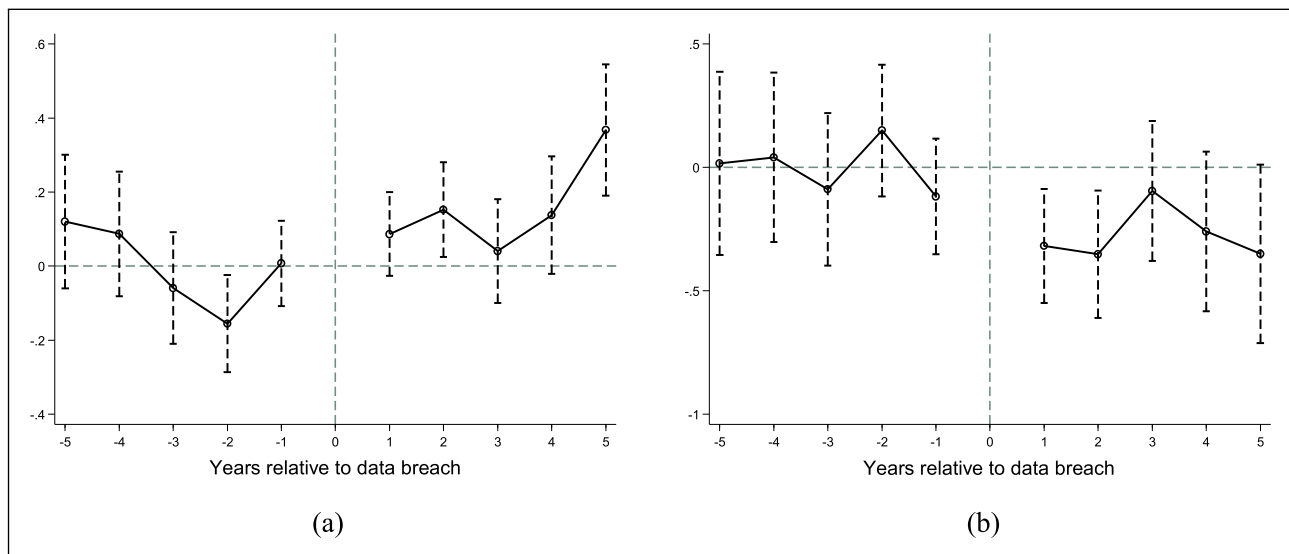


Figure 4. Parallel trends plot: (a) dynamic trend of the IT investment intensity; (b) dynamic trend of a new IT investment.

5.4.1 Testing for Parallel Pre-Trends. To ensure the validity of the DiD estimates, satisfying the parallel trends assumption is crucial. The assumption of parallel pre-trends implies that, in the pre-breach stage, the trends of either outcome variable of interest (*Intensity* or *Innovation*) would follow similar trajectories across treatment and control groups. Significant differences in trends would violate the parallel trends assumption, indicating potential factors that may confound the causal interpretation. A common test for this assumption is the dynamic DiD test, involving an examination of how the impact of data breaches varies over time and whether the trends of the outcome variables are parallel across groups in the pre-treatment period. In this study, the test is conducted using the

following model equation:

$$\text{Outcome}_{i,j} = \beta_0 + \sum_{-5 \leq t \leq 5} \beta_t \text{Year}_{i,j}^t + \sum \alpha \text{Control}_{i,j} + v_i + w_j + \epsilon_{ij}. \quad (2)$$

In Equation (2), t represents the number of years relative to the data breach year (e.g., if a firm experienced a data breach in 2013, then for 2012, $t = -1$, and for 2014, $t = 1$). *Year* is a time indicator, taking a value of 1 if the year corresponds to t years relative to the firm's first data breach. To avoid multicollinearity, we followed the standard practice of dynamic DiD and exclude the data breach year (i.e., Year^0) (e.g., Baker et al., 2022; Zhang et al., 2023).

Building on the detrending process (Beck et al., 2010), Figure 4 presents the estimated β_t values with 90% confidence intervals plotted against the number of years relative to the breach year. In Figure 4, column (a) shows that for the years ranging from -5 to -1 (i.e., the pre-breach years), the point estimators of β_t are clustered around the zero-horizontal line, indicating that firms' IT investment intensity does not exhibit any significant pre-trend. Moreover, the coefficients of β_t for all pre-breach years are generally insignificant, with the corresponding 90% confidence intervals largely encompassing both positive and negative values. Therefore, this evidence suggests that there is no significant difference in the pre-existing trends in IT investment intensity between the treatment group and the PSM-matched control group.

In Figure 4, column (b) illustrates that during the pre-breach years ranging from -5 to -1 , the point estimates of β_t cluster around the zero-horizontal line, and the corresponding 90% confidence intervals encompass both positive and negative values. Therefore, the aforementioned evidence suggests that there is no significant difference in the pre-existing trends in the new IT investment aspect between the treatment group and the PSM-matched control group.

Recognizing the critical importance of satisfying the parallel trends assumption in the DiD estimations, we present supplementary evidence to support this assumption. We followed recent studies (Khern-am-nuai et al., 2018; Liu et al., 2023; Pamuru et al., 2021) and performed Augmented Dickey–Fuller (ADF) tests and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) tests. The outcomes of these tests provide additional evidence to support the fulfillment of the parallel trends assumption in our analyses, indicating no significant differences between the treatment and control groups in the pre-treatment period for either outcome variable (*Intensity* or *Innovation*). For a more detailed description of these tests and their results, please refer to the E-companion (EC-7).

5.4.2 Alternative Matching Approaches. PSM will result in a loss of sample data, potentially undermining the results' persuasiveness. Additionally, using a single-matching approach may introduce bias. To address these concerns, we alternatively employed entropy balancing, a novel multivariate matching method introduced by Hainmueller (2012) and extensively used in recent research (e.g., Hendricks et al., 2019; Zhang et al., 2023). This method adjusts sample weights to create a control group with covariate distributions similar to the treatment group without reducing the sample size. We replaced PSM with entropy balancing to achieve covariate balance in our matched samples.¹⁶ The balancing test results, presented in E-companion (EC-8), support the effectiveness of the entropy balancing method. We replicated our baseline analyses (regarding H1 to H4) using the entropy balancing-matched sample. The results in Table 3 show that for both main and moderating effects, the findings are consistent with the baseline analyses, reinforcing the robustness of our results.

Table 3. Entropy balancing matching.

Dependent variable:	<i>Intensity</i>	<i>Innovation</i>	<i>Intensity</i>	<i>Innovation</i>
For testing	H1	H2	H3	H4
	(1)	(2)	(3)	(4)
<i>Post breach</i>	0.234*** (5.895)	−0.132* (−1.744)	0.004 (0.062)	−0.415*** (−2.949)
<i>Post breach × quality culture</i>			0.078*** (3.268)	
<i>Post breach × innovation culture</i>				0.053*** (2.720)
<i>Quality culture</i>			0.032* (1.925)	
<i>Innovation culture</i>				−0.039* (−1.856)
<i>Intercept</i>	Yes	Yes	Yes	Yes
<i>Controls</i>	Yes	Yes	Yes	Yes
<i>Firm fixed effects</i>	Yes	Yes	Yes	Yes
<i>Year fixed effects</i>	Yes	Yes	Yes	Yes
<i>N of observations</i>	17,319	14,584	11,675	10,574
<i>Adjusted R²</i>	0.883	0.112	0.881	0.100

Note: This table presents coefficients obtained from DiD regressions using entropy balancing. All specifications remain consistent with the baseline analyses, while alternatively employing the entropy balancing-matched sample. The t-statistics are reported in parentheses. The symbols ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Furthermore, to address potential bias from PSM design choices, we employed two alternative PSM designs. Detailed results in E-companion (EC-9) show that both alternatives yield consistent results with our baseline analyses for the main and moderating effects, further supporting the robustness of our findings. Additionally, we performed placebo tests to mitigate biases from chance or other latent time series factors (E-companion EC-10), a stacked DiD approach to minimize estimation biases arising from staggered adoption (E-companion EC-11), and a two-step Heckman correction to further address the concern that a firm's IT investment information might not be randomly covered by the CI technology database (E-companion EC-12).

6 Post-Hoc Investigation

6.1 Does the Scale of Data Breaches Matter?

In the following sections, we delved deeper into the heterogeneity of treatment effects to further explore potential generalizability. We first investigated the role of the breach scale in shaping the effects of data breaches. The breach scale, defined as the number of records affected by the data breach, plays a crucial role in determining the magnitude of the breach's impact. Larger-scale data breaches often result in more severe consequences, such as higher costs (Ponemon Institute, 2014) and greater customer confidence loss (Kwon and Johnson, 2015). These severe consequences can have far-reaching implications for a firm, including damage to its reputation, erosion of customer trust, and increased regulatory scrutiny (Gwebu et al., 2018; Janakiraman et al., 2018). Given

the substantial repercussions of larger breaches, the management becomes more attentive and faces mounting pressure to address the fallout. This increased scrutiny prompts managers to exhibit a more pronounced response to data breaches. Considering that both the promoting and inhibiting effects of data breaches represent certain forms of firms' post-breach responses, we predicted that the dual impact of data breaches on IT investments, both promoting and inhibiting, will be enhanced following large-scale data breaches.

To test the aforementioned prediction, we conducted a stratified analysis of the effects of data breaches across different breach scales, similar to previous studies (e.g., Dai et al., 2020; Wang et al., 2024). We constructed a series of treatment indicators similar to *Post Breach*, each corresponding to a different scale of data breach. For example, *Post Breach_1000* is a treatment indicator that equals 1 for the years following the first data breach with a scale of less than 1,000 records and 0 otherwise. Subsequently, we conducted impact analyses for each breach scale and explored the effect heterogeneity across various breach scales.

Table 4 shows the results of the aforementioned stratified analysis. We observed that the impact of data breaches on the IT investment intensity is contingent on the scale of the breach. The results in columns (1)–(5) suggest that in cases where the breach scale is below approximately 1,000, the impact on IT investment intensity seems to diminish. On the other hand, the results in columns (6)–(10) imply that the effect of data breaches on new IT investments also hinges on the scale of the breach, and for breaches below approximately 10,000, the impact diminishes. Overall, we found that the dual effects of data breaches on IT investment are strengthened following large-scale breaches.

6.2 How Does the Innovative Nature of New IT Investments Matter?

Furthermore, we expected that the impact of data breaches on reducing new IT investments will vary depending on the innovativeness of the IT solutions involved. Innovativeness, in this context, refers to the degree to which an IT solution is novel, emerging, and demonstrates forward-thinking impacts (Fichman and Kemerer, 1997; Jacobs et al., 2015; Lyytinen and Rose, 2003). When firms adopt IT solutions that are considered “new to the world,” they often face additional knowledge barriers and operational threats due to the unproven and nascent nature of the ITs (Jacobs et al., 2015; Lyytinen and Rose, 2003). Consequently, the adoption of these emerging technologies typically results in significant threats and uncertainties in operations. “Threat rigidity” fundamentally arises from firms' attempts to mitigate unknown threats and uncertainties by exhibiting rigidity and conservatism. Hence, the considerable threats and uncertainties associated with innovative new IT solutions are likely to intensify breached firms' inclination toward “threat rigidity.” Thus, we predicted that the reduction in new IT investments post-breach, driven by the

threat rigidity mechanism, will be more pronounced if the IT solutions involved are highly innovative.

To examine the varied treatment effects based on the innovativeness of new IT investments, we used the criteria established by Wang et al. (2023) to classify each IT in the CI technology database as either innovative or non-innovative.¹⁷ In their classification, the fundamental distinction between these two groups lies in whether IT is considered new to the world and represents an emerging technology, aligning with the scenario we are interested in. The detailed classification list can be found in Table EC-2 (E-companion). Based on this categorization, we constructed *Innovative Innovation* and *Non-Innovative Innovation*. Both variables are constructed in a manner similar to the *Innovation* variable, with the only difference being that they specifically include ITs classified as either innovative or non-innovative.

Then, we repeated the PSM-combined-DiD analyses in Table 1, alternatively using *Innovative Innovation* or *Non-Innovative Innovation* as the outcome variables. The results in Table 5 show that across different specifications, the coefficient of *Post Breach* is significant only when *Innovative Innovation* is the outcome variable. In contrast, when the outcome variable is *Non-Innovative Innovation*, the coefficient of *Post Breach* is insignificant across different model specifications. This indicates that the effect of data breaches on decreasing new IT investments applies only to innovative IT investments in new domains, thereby supporting our predication.

6.3 How Does the Security Orientation of New IT Investments Matter?

The impact of data breaches on a firm's new IT investments may also depend on whether the new IT investments are security-oriented. IT applications with a security focus, such as vulnerability management and data backup, are typically designed to mitigate potential security risks (Angst et al., 2017). Consequently, firms generally perceive these IT applications as secure and less likely to pose security threats to their systems. Therefore, breached firms may exhibit less concern about potential security threats when adopting such new ITs, showing a lower tendency toward threat rigidity. Consequently, we anticipated that the decrease in new IT investments following a breach, underpinned by the threat rigidity mechanism, will be less significant if the IT solutions are security-oriented.

To examine this, we constructed separate measures for non-security new IT investments (*Non-Security Innovation*) and security new IT investments (*Security Innovation*). Specifically, we classified IT applications used in new IT investments into non-security (e.g., business intelligence, enterprise resource planning) and security (e.g., firewall, security software) categories based on their functionality. A detailed list of these applications is provided in Table EC-2 (E-companion). Similar to measuring new IT investment (*Innovation*), we operationalized *Non-Security Innovation* and *Security Innovation*.

Table 4. Analyzing the stratified effects of data breach using the breach scale.

Dependent variable	Intensity					Innovation				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>Post Breach_100</i>	0.088 (1.54)					−0.007 (−0.06)				
<i>Post Breach_1000</i>		0.108* (1.95)					−0.060 (−0.53)			
<i>Post Breach_10000</i>			0.092* (1.79)					−0.143 (−1.38)		
<i>Post Breach_100000</i>				0.104** (2.12)					−0.203** (−2.05)	
<i>Post Breach_1000000</i>					0.119** (2.46)					−0.182* (−1.86)
<i>Intercept</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Firm fixed effects</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Year fixed effects</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>N of observations</i>	3,411	3,411	3,411	3,411	3,411	3,200	3,200	3,200	3,200	3,200
<i>Adjusted R²</i>	0.872	0.872	0.872	0.872	0.872	0.121	0.121	0.122	0.123	0.122

Note: This table presents the coefficients obtained from DiD regressions that investigate the impact of data breaches on IT investment across various breach scales. We constructed a set of treatment dummies similar to *Post Breach* based on the scale of the data breach. All other specifications remained consistent with the baseline analysis. The *t*-statistics are reported in parentheses. The symbols ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table 5. Differences in the effects of data breach on innovative/non-innovative new IT investment.

Dependent variable	Innovative Innovation		Non-Innovative Innovation	
	(1)	(2)	(3)	(4)
<i>Post breach</i>	−0.119** (−2.332)	−0.125** (−2.453)	−0.077 (−1.069)	−0.095 (−1.320)
<i>Intercept</i>	Yes	Yes	Yes	Yes
<i>Controls</i>	No	Yes	No	Yes
<i>Firm fixed effects</i>	Yes	Yes	Yes	Yes
<i>Year fixed effects</i>	Yes	Yes	Yes	Yes
<i>N of observations</i>	3,200	3,200	3,200	3,200
<i>Adjusted R²</i>	0.163	0.167	0.063	0.072

Note: This table provides coefficients from DiD regressions that investigate treatment heterogeneity across innovative and non-innovative new IT investments. All other specifications remain consistent with the baseline analysis in Table 1, alternatively using *Innovative Innovation* or *Non-Innovative Innovation* as the outcome variable. The *t*-statistics are reported in parentheses. The symbols ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table 6 presents the results of the DiD regression analysis. The results show that, across various specifications, the coefficient for *Post Breach* is significant only when *Non-Security Innovation* serves as the outcome variable. In contrast, when the outcome variable is *Security Innovation*, the coefficient for *Post Breach* is consistently insignificant across different model specifications. These findings suggest that data breaches primarily decrease investments in non-security IT categories,

Table 6. Differences in the effects of data breach on non-security/security new IT investment.

Dependent variable	Non-Security Innovation		Security Innovation	
	(1)	(2)	(3)	(4)
<i>Post breach</i>	−0.168** (−2.064)	−0.185** (−2.272)	−0.027 (−0.758)	−0.035 (−0.966)
<i>Intercept</i>	Yes	Yes	Yes	Yes
<i>Controls</i>	No	Yes	No	Yes
<i>Firm fixed effects</i>	Yes	Yes	Yes	Yes
<i>Year fixed effects</i>	Yes	Yes	Yes	Yes
<i>N of observations</i>	3,200	3,200	3,200	3,200
<i>Adjusted R²</i>	0.118	0.127	0.071	0.076

Note: This table provides coefficients from DiD regressions that investigate treatment heterogeneity across security and non-security new IT investments. All other specifications remain consistent with the baseline analysis, alternatively using *Non-Security Innovation* or *Security Innovation* as the outcome variable. The *t*-statistics are reported in parentheses. The symbols ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

while investments in security IT categories remain relatively unaffected. This provides support for our prediction.

7 Concluding Remarks

This study offers valuable insights into firms' responses to data breaches by reconciling seemingly competing perspectives of "failure learning" and "threat rigidity" within a unified framework. Utilizing a sample of data breaches spanning from 2012 to 2018 and employing a PSM-combined-DiD approach, we

uncovered several key findings. In our primary analyses, we observed that data breaches exert dual effects on the firms' IT investments. On the one hand, they enhance the intensity of IT investments (a promoting effect), while concurrently reducing new IT investments (an inhibiting effect). Moreover, we found that these dual effects of data breaches are contingent upon organizational culture, quality culture amplifies the promoting effect, and innovative culture alleviates the inhibiting effect. Subsequently, in post-hoc analyses, we found that the dual impact of data breaches, both promoting and inhibiting effects, depends upon the scale of the breaches. Furthermore, we found that data breaches exclusively inhibit investments in innovative or non-security-oriented new ITs, while exhibiting no significant impact on non-innovative or security-oriented ones.

7.1 Theoretical Implications

This study makes significant contributions to academic literature in several key ways. First, our study contributes to the data breach literature by shedding light on the dual impact of data breaches on firm IT investments—both promoting and inhibiting effects. While prior research has extensively explored firms' responses to data breaches from the perspective of "failure learning," such as enhancing IT capabilities (Benaroch and Chernobai, 2017), engaging in more CSR activities (Lending et al., 2018), or undertaking unrelated asset divestitures as remedial measures post-breach (Say and Vasudeva, 2020), another important failure-response perspective of "threat rigidity" has been largely overlooked. Our study fills this gap by revealing that firms' responses to data breaches can also manifest in a "threat rigidity" manner, offering a novel perspective on understanding the impact of data breaches. By identifying and empirically demonstrating the concurrent existence of the dual impacts of data breaches, our research contributes to a more comprehensive understanding of how organizations respond to and manage the aftermath of data breaches.

Second, as big data becomes more prevalent, the focus on information security has intensified in operations management (OM), garnering increasing interest from scholars in the field (Bensoussan et al., 2020; Cezar et al., 2017; Luo and Choi, 2022; Massimino et al., 2017; Massimino et al., 2018; Mookerjee and Samuel, 2023). However, despite the critical importance of information security, OM scholars have not yet devoted sufficient attention to this area. Massimino et al. (2018) underscored this research gap by noting that "digital confidentiality has received significantly less research attention from operations and supply chain management (OSCM) scholars compared to cost, quality, flexibility, delivery, and innovation" (p. 1492). Our study contributes meaningfully to filling these gaps in the current OM literature by further examining the effects of data breaches on firm operations. Moreover, our study provides a timely response to the recent research call by Kumar and Mallipeddi (2022) for expanded investigations into cybersecurity from the OM standpoint.

Third, this study provides groundbreaking insights by showing that both "failure learning" and "threat rigidity" can coexist within organizational responses to failure. Prior research has often portrayed these two mechanisms as mutually exclusive, with one dominating over the other based on factors such as the magnitude of potential losses or the availability of organizational resources. For instance, "failure learning" tends to prevail when losses are substantial (Laughunn et al., 1980; March and Shapira, 1987; Shimizu, 2007), whereas "threat rigidity" emerges as the dominant response in resource-constrained environments (Audia and Greve, 2006; Shimizu, 2007). However, existing studies have overlooked the possibility of these two mechanisms coexisting within different aspects of the same strategy. Our research breaks new ground by providing empirical evidence of this coexistence, thereby offering a deeper understanding of how organizations respond to failure. By revealing that both "failure learning" and "threat rigidity" can operate simultaneously, our study enriches the scholarly discourse on organizational responses to failure and opens up avenues for further exploration in this area.

Fourth, this study extends the existing literature on IT investments. Previous research has primarily focused on examining the influence of internal or external characteristics on firms' IT investments (e.g., Xue et al., 2017; Xue et al., 2008). However, there has been limited empirical exploration into how specific firm events or failures affect these investments. Our study helps to fill this gap by highlighting the importance of firm failures, such as data breaches, in shaping their IT investment decisions. This focus on the impact of specific events enriches our understanding of the dynamics between firm failures and their IT investment strategies, contributing to the broader literature on IT investment decision-making.

Fifth, our study has identified a series of contextual factors that influence firms' responses to data breaches. We illuminated the pivotal role of corporate culture in shaping how firms respond to such breaches. Additionally, our findings indicate that the scale of data breaches and the characteristics of IT solutions, such as innovativeness and security orientation, significantly impact firms' post-breach responses in terms of IT investment. These insights provide a more nuanced understanding of how firms adjust their IT investments following a breach.

7.2 Practical Implications

Given the pervasive use of data in businesses, understanding the consequences of data breaches on firms' strategic decision-making has become increasingly critical. Our research sheds light on a dual trend in IT investments observed in firms after experiencing data breaches: on the one hand, firms take proactive measures to "enhance" their security posture, evident in increased investments across the entire IT domain; on the other hand, they also adopt cautious and protective measures to safeguard their assets, displaying restraint in IT innovation. These

insights offer a better understanding of how firms strategize in the aftermath of data breaches.

Moreover, our findings align with the well-established notion of IT being a double-edged sword for information security (Ransbotham et al., 2016). Specifically, on the bright side, IT investments can bolster firms' security by enhancing their IT capabilities. Conversely, IT also harbors potential risks, creating dark sides such as significant security vulnerabilities in firms' operations. Our research demonstrates that firms, post-data breach, will carefully weigh the dual impact of IT on security and implement tailored measures accordingly to strike the right balance.

7.3 Limitations and Future Research

Like other studies, our study has certain limitations that present opportunities for future investigation. First, due to constraints in available data, we were unable to further differentiate IT investment intensity based on innovativeness or security orientation. This limitation hinders our exploration of whether the strength of the "failure learning" trend after data breaches depends on such categorization. Future research could address this limitation by utilizing alternative data sources or conducting surveys to classify IT investments and investigate their implications. Second, our empirical analysis focuses exclusively on US firms, potentially limiting the generalizability of the findings. To enhance the robustness and applicability of the results, future research could conduct cross-country studies to examine how data breaches impact IT investment in different cultural, regulatory, and economic contexts. Third, this study primarily examines post-breach responses within firms. Future research could explore whether the impact of data breaches extends beyond individual firms and whether it generates any spillover effects on peers of the breached firms.

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Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Notes

1. <https://www.idtheftcenter.org/post/2023-annual-data-breach-report-reveals-record-number-of-compromises-72-percent-increase-over-previous-high/>
2. <https://www.ibm.com/downloads/cas/3R8N1DZJ>
3. <https://www.securityweek.com/target-provides-update-costs-related-data-breach/>
4. <https://www.complianceweek.com/data-privacy/study-fear-of-data-breaches-stifling-bank-innovation/28634.article>
5. <https://www.pwc.com/gx/en/ceo-survey/2020/reports/pwc-23rd-global-ceo-survey.pdf>
6. <https://www.ibm.com/downloads/cas/RDEQK07R>
7. <https://wrds-www.wharton.upenn.edu/pages/get-data/audit-analytics/accounting-oversight/cybersecurity/>
8. <https://privacyrights.org/categories/data-breaches>
9. <https://www.idtheftcenter.org/notified/>
10. For breached firms sourced from WRDS, we directly used the stock ticker information provided by that source. For each breached firm sourced from PRC or ITRC, we manually matched the breached firm's name with the firm names in Compustat to identify their stock ticker information.
11. The CI technology database provides information on IT implementation starting from 2011. Moreover, the identification of new IT adoptions involves a two-year window. Therefore, our sample starts in 2012. However, our data end in 2018 due to the constraints of the CI technology database we acquired.
12. We excluded the IT applications related to printers and phones because these ITs are rarely associated with information security risks.
13. We appreciate the authors of this study for publicly sharing the data on corporate culture scores.
14. In Model 2, Table 1, the coefficient for *Post Breach* is 0.090. This indicates that when the *Post Breach* variable changes from 0 to 1, the natural logarithm of the firm's new IT investment level is $e^{0.090}$ times the original level, corresponding to an increase by $(e^{0.090} - 1) \times 100\% = 9.4\%$.
15. In Model 4, Table 1, the coefficient for *Post Breach* is -0.202 . From Table EC-6 (E-companion), we can see that the mean of the *Innovation* is 0.533. Then, $0.202/0.533 = 37.9\%$.
16. *Firm Size*, *Firm Age*, *Firm ROA*, and *Digital Connectivity Scope* are specified as covariates to be balanced using entropy balancing.
17. To categorize each IT in the CI technology database as either innovative or non-innovative, Wang et al. (2023) initially classified 23 types of IT that had widespread use before the 1990s as non-innovative. Following this, they invited 60 professionals,

including chief information security officers (CISOs), cybersecurity risk directors, IT development directors, and risk consulting experts, to participate in a survey evaluating the innovativeness of each remaining type of IT. The evaluations provided by these experts are then used by the authors to classify each of the remaining types of IT as innovative or non-innovative.

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